

vLLM P/D Disaggregation and Mooncake integration

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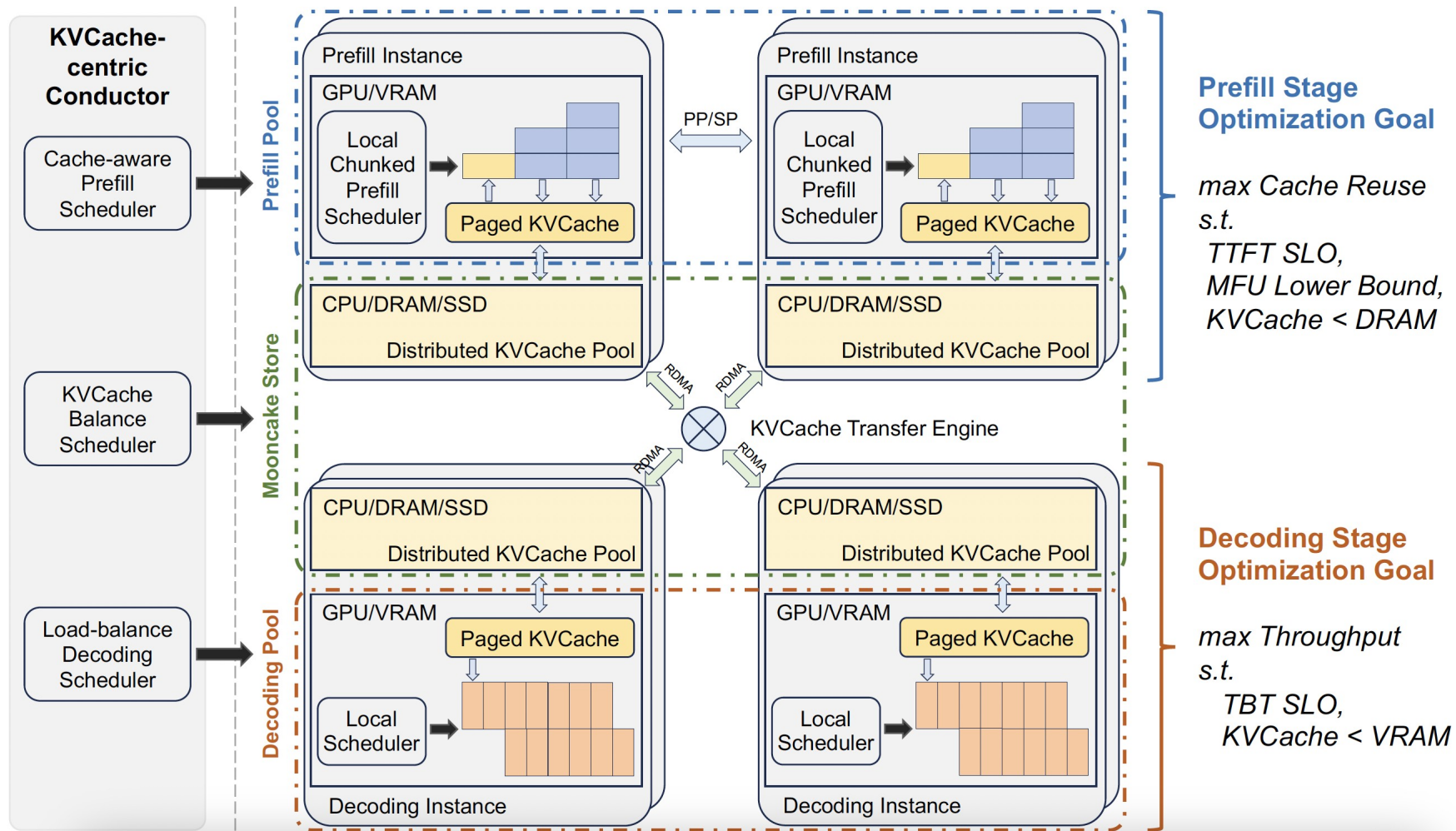


Part 01

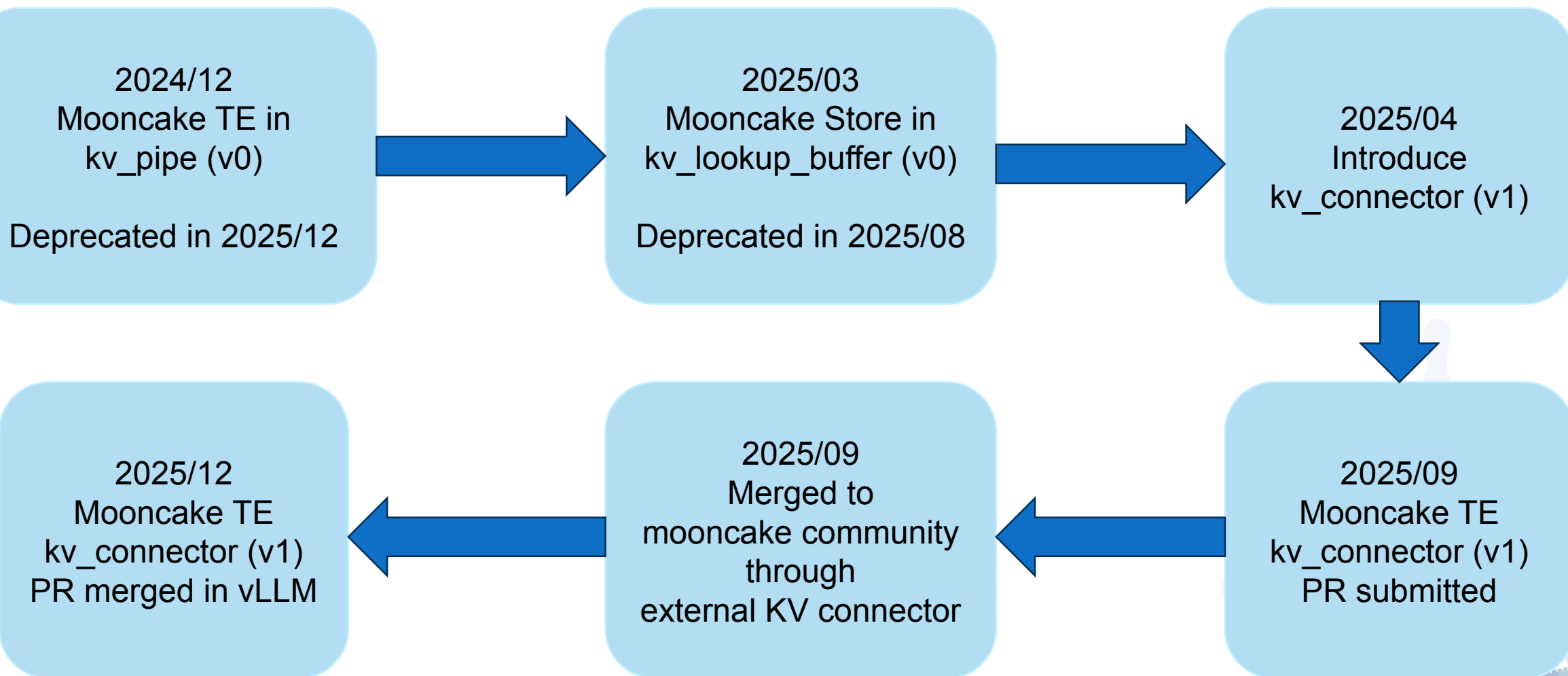
Background



What's Mooncake



vLLM kv connector history



Part 02

KV connector detail

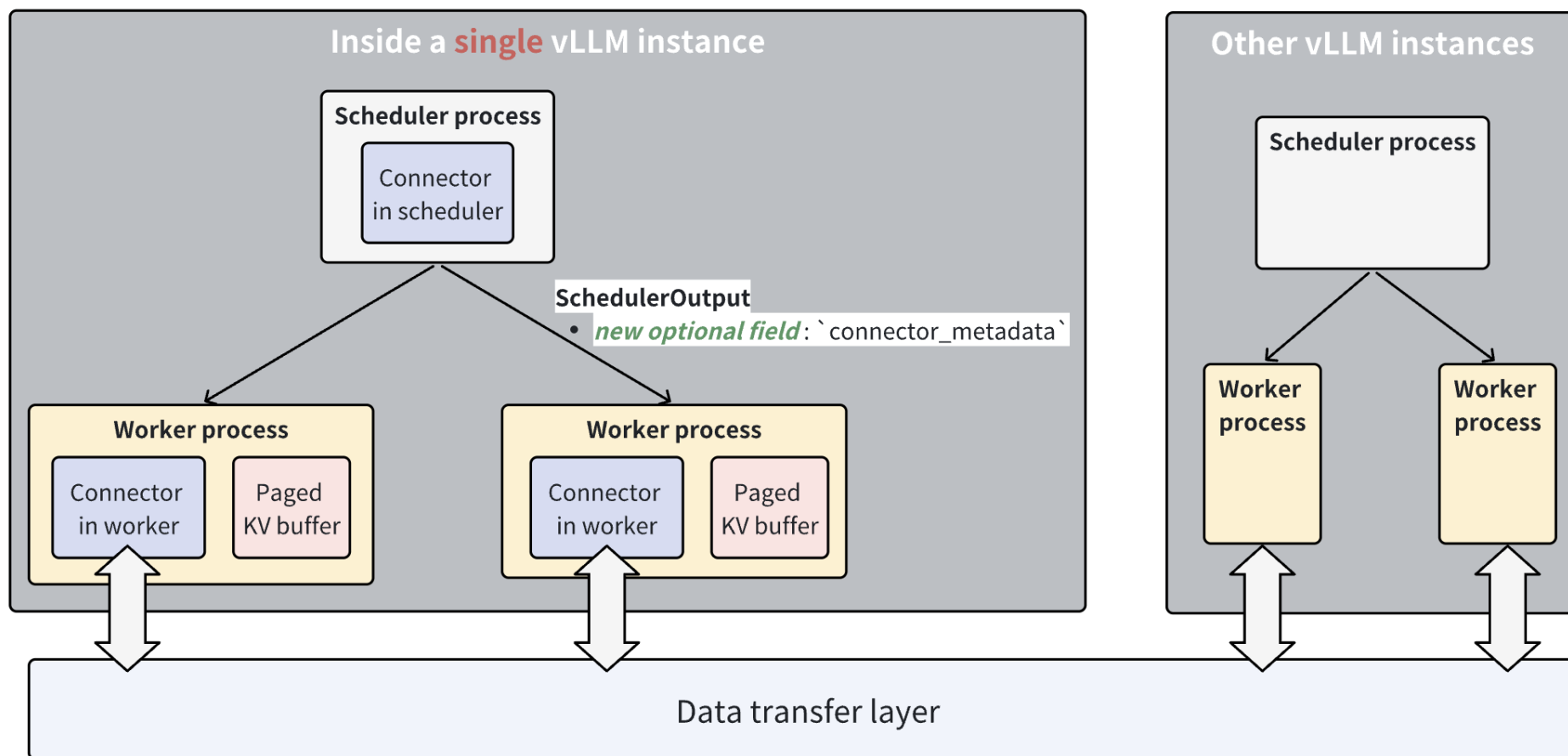


vLLM KV cache transfer abstractions

- KV connector: The most powerful and flexible layer. Provides the highest degree of freedom by allowing direct intervention into the control flow of the vLLM scheduler and workers.
- KV pipe (only in v0, deprecated in v1): Provide a simple mechanism for data transmission. Historically, it was designed to be easily adaptable to various transport backends, such as Mooncake transfer engine.
- KV lookup buffer (only in v0, deprecated in v1): Retrieve the KV cache for specific tokens using a key-value indexing relationship. Historically, it could be easily integrated with key-value systems like Mooncake store.

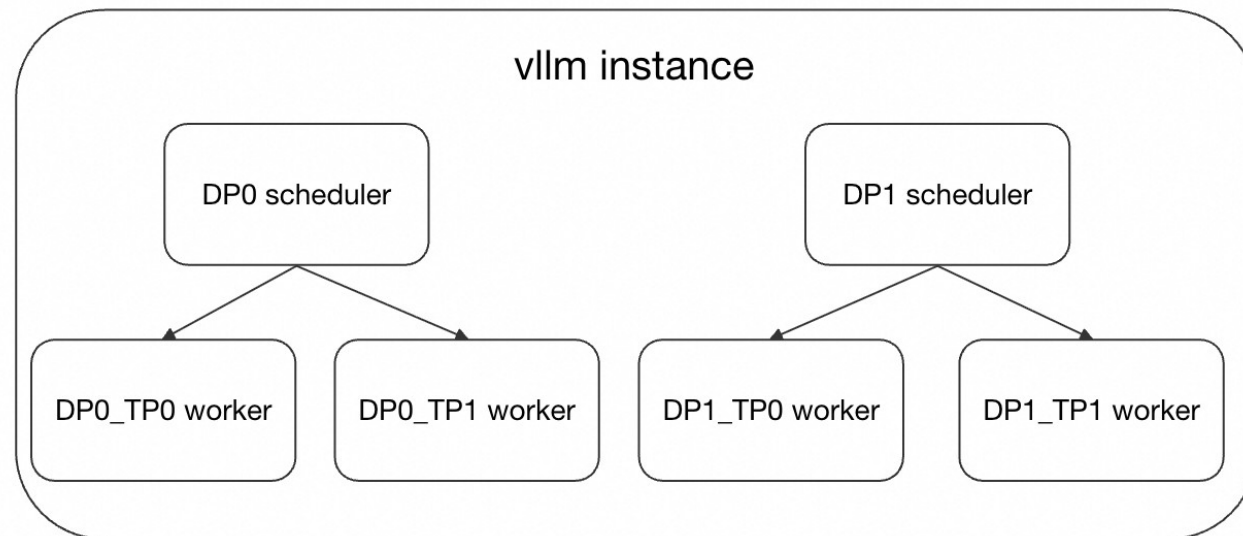


vLLM KV connector

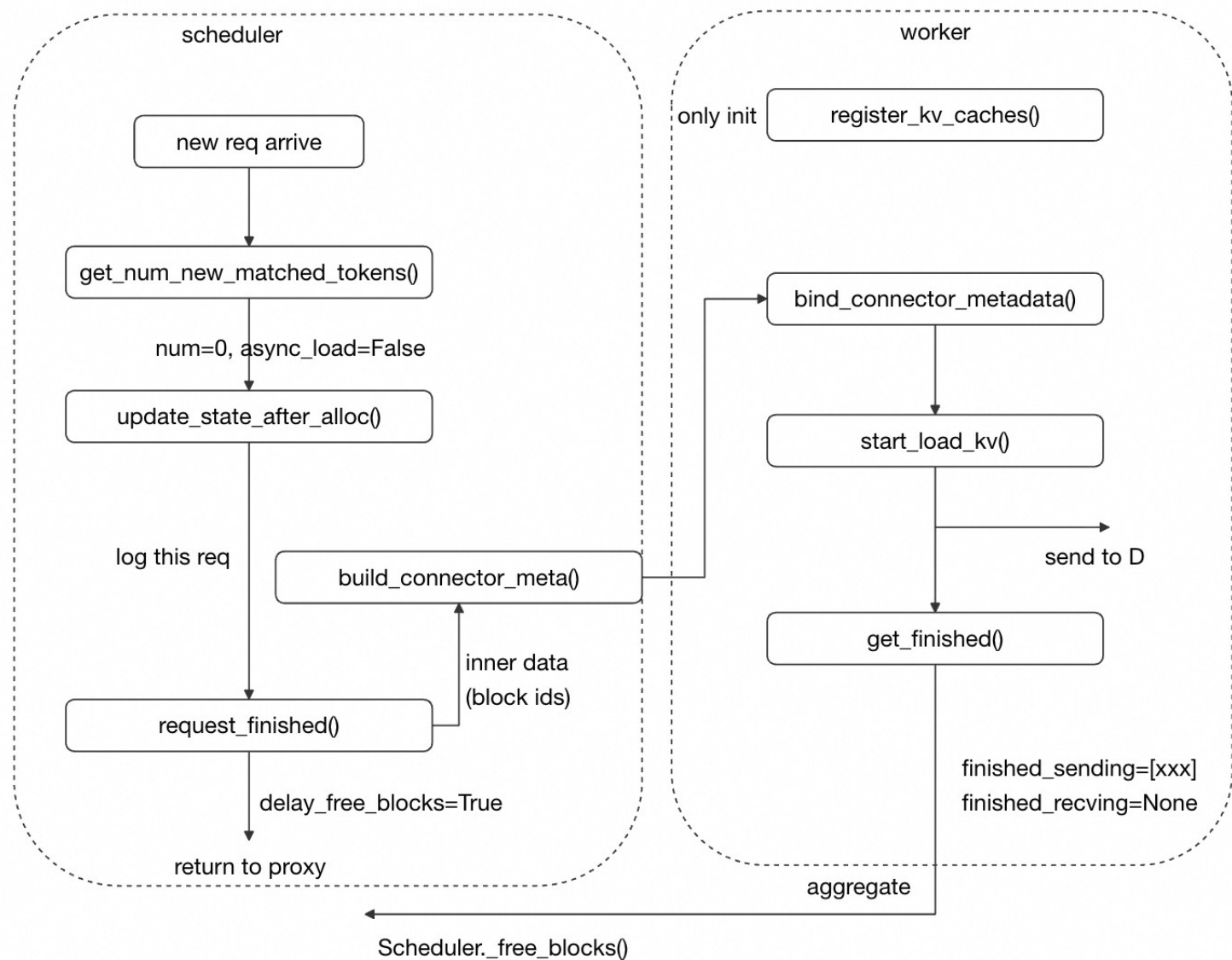


vLLM KV connector

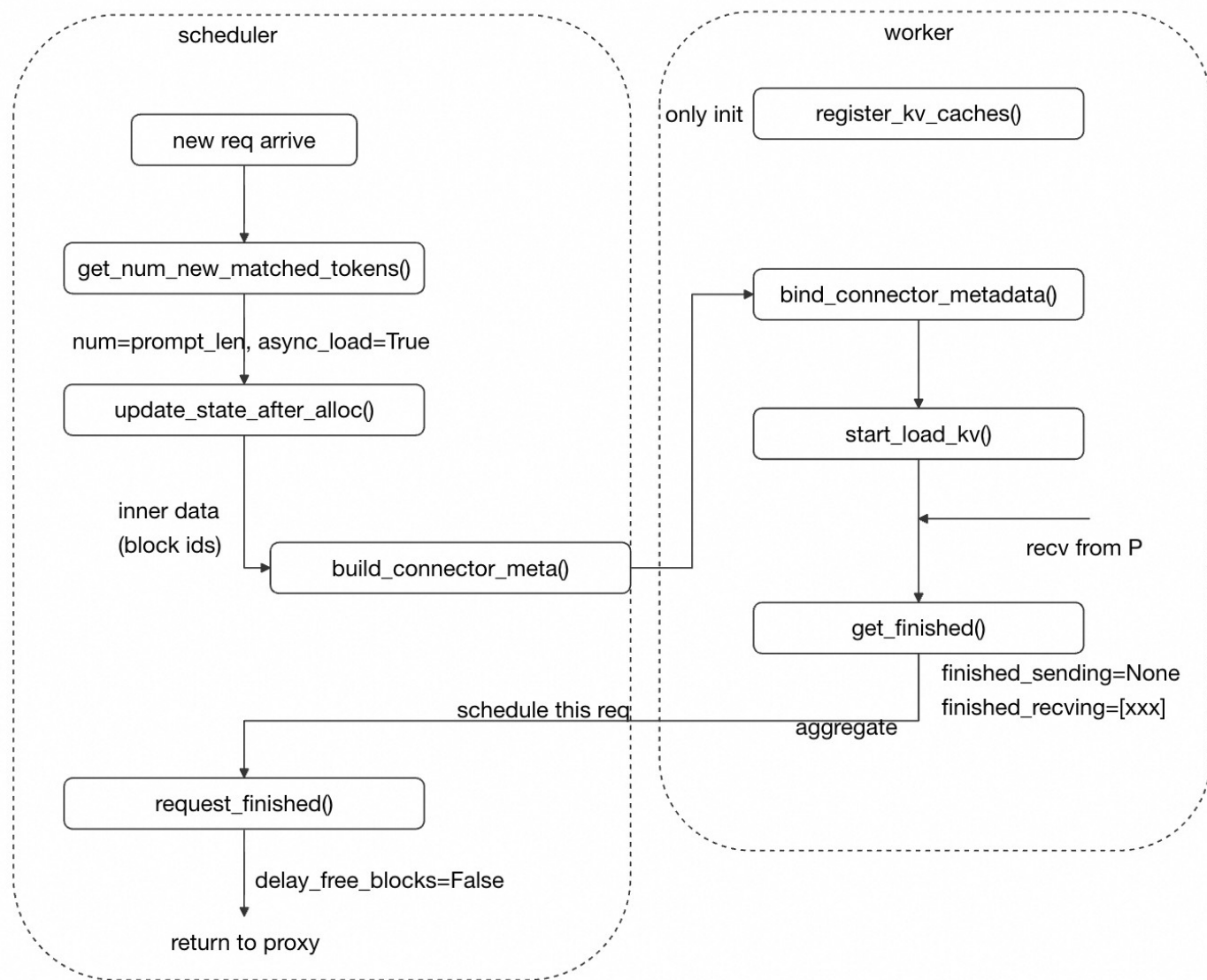
- connector scheduler: Responsible for making decisions related to KV cache memory allocation and deallocation.
- connector worker: The actual worker instance that performs the data transfer.



vLLM KV connector: prefill example



vLLM KV connector: decode example

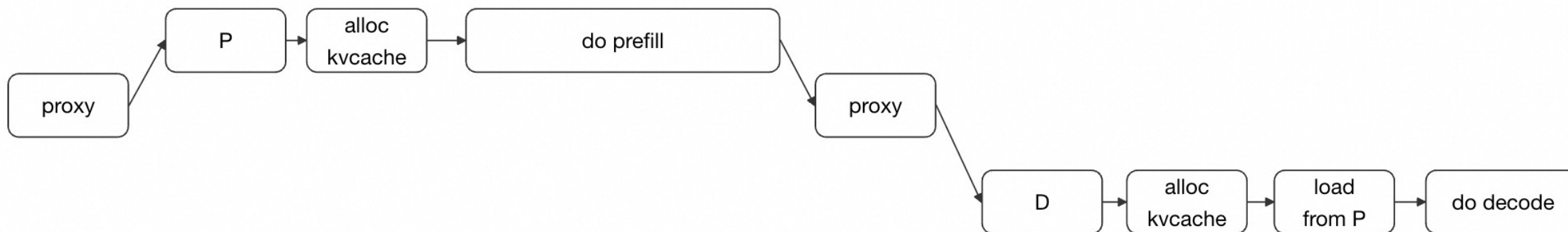


Part 03 Mobile integration

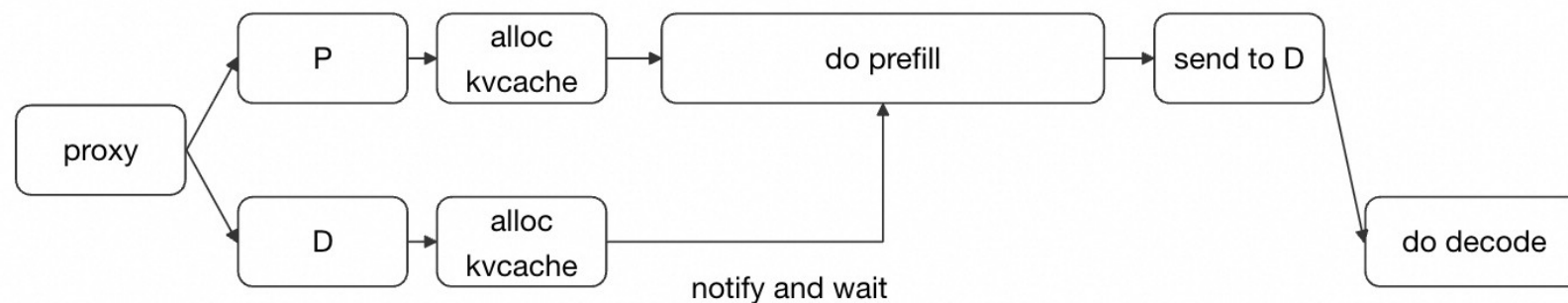


Proxy mode

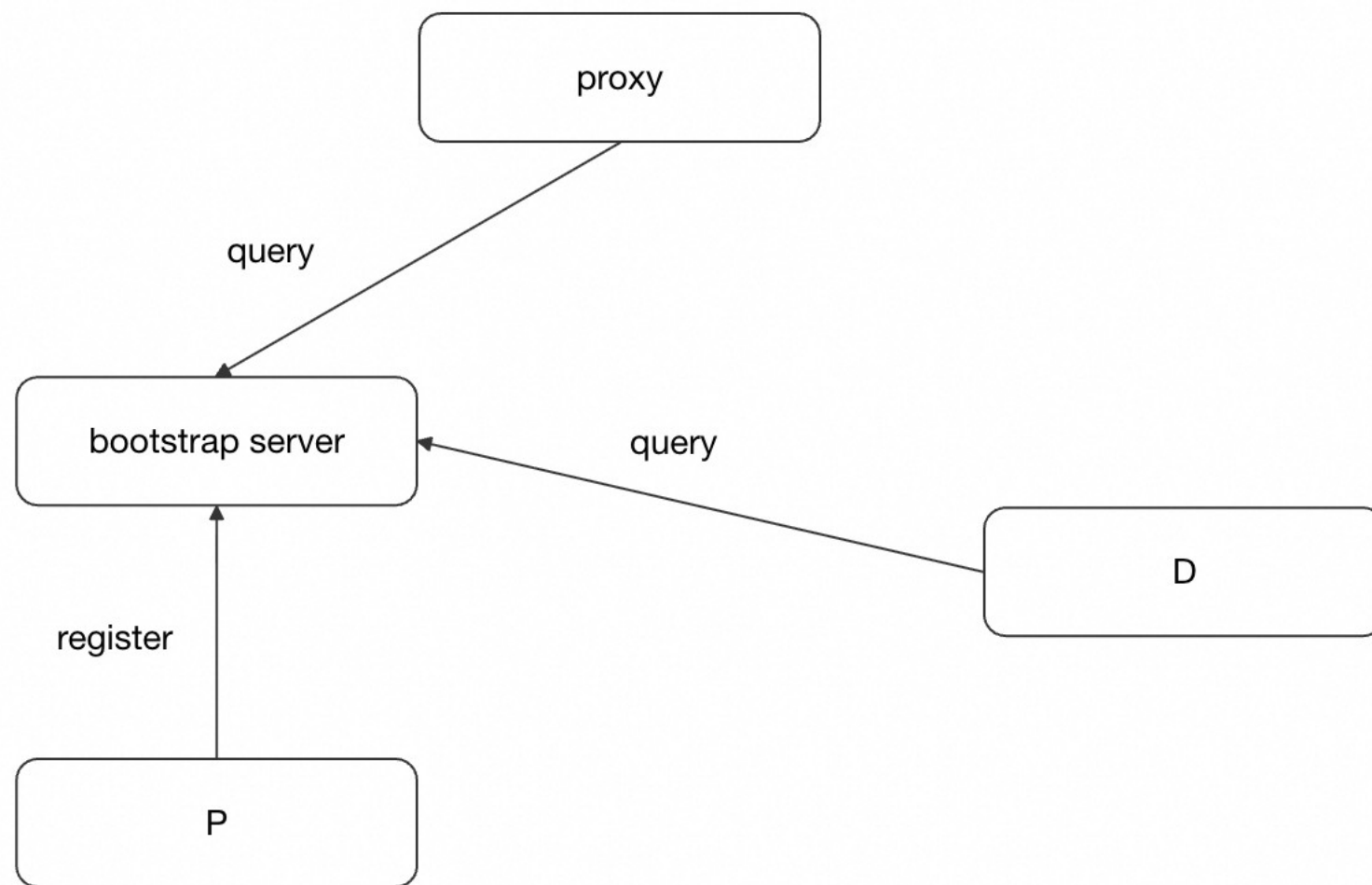
Used by most vLLM KV connectors. (e.g., NixlConnector)



Used by MooncakeConnector and MoRIIOConnector (WRITE MODE)



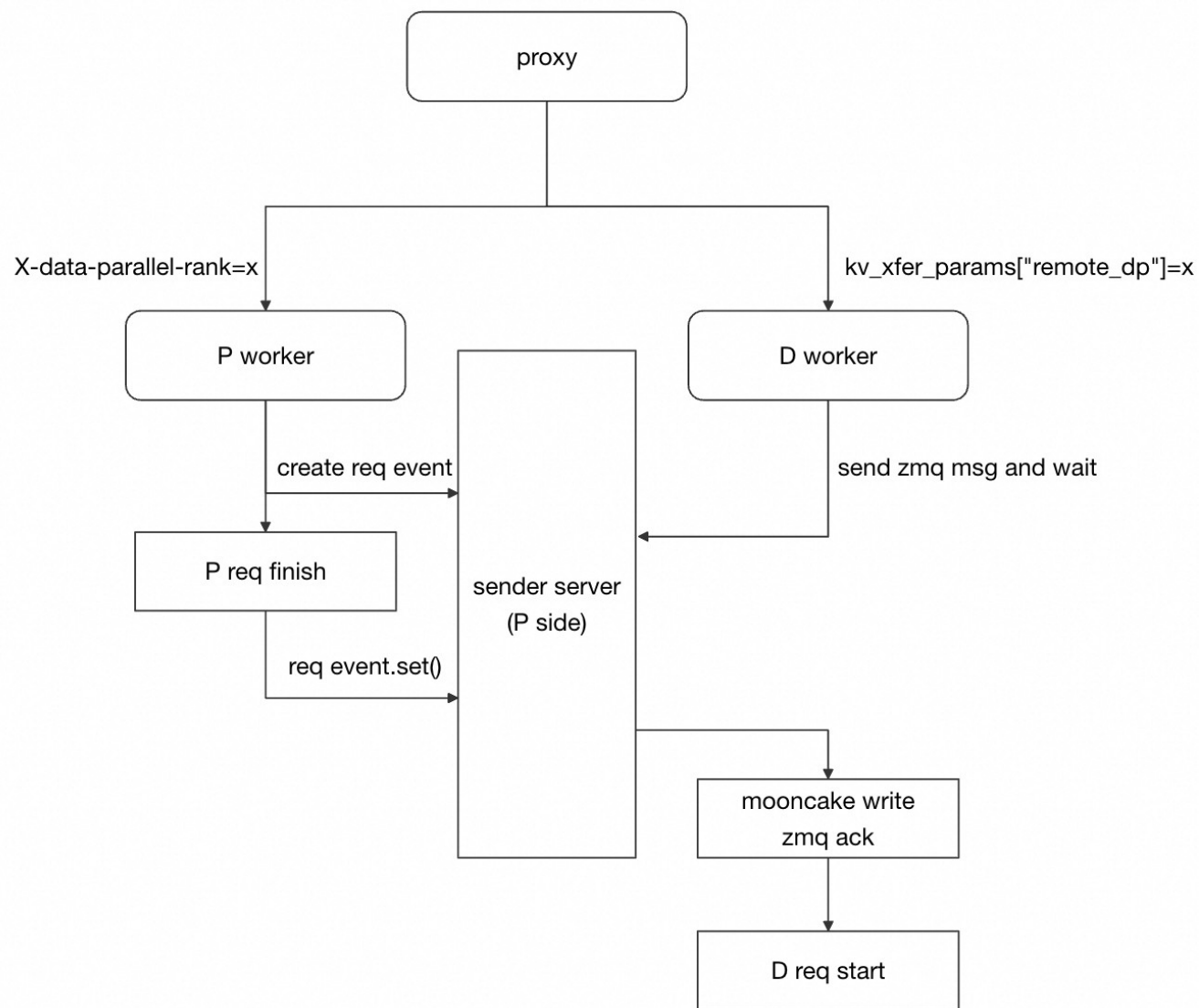
MooncakeConnector workflow



central bootstrap server:
log per DP/TP/TP worker



MooncakeConnector workflow



Optimization:

- group reqs in zmq msg (D side)
- async sender server with thread poll
- wait for first event.set() reqs, send and ack partly of zmq msg
- group contiguous blocks and use mooncake.batch_transfer_sync_write



Work with k8s

```
1 # Joint Solution: RBG + vLLM + Mooncake Connector
2 apiVersion: workloads.x-k8s.io/v1alpha1
3 kind: RoleBasedGroup
4 metadata:
5   name: vllm-pd-with-mooncake-demo
6 spec:
7   roles:
8     # 1. Gateway: Routing to vLLM instances (SMG or vLLM Proxy)
9     - name: proxy
10      dependencies: [ "prefill", "decode" ]
11      replicas: 1
12      template:
13        spec:
14          containers:
15            - name: proxy
16              image: lightseekorg/smg:latest
17              command:
18                - smg
19                - --prefiller-host
20                - http://vllm-pd-with-mooncake-demo-prefill-0.s-vllm-pd-with-
21                - --prefiller-port
22                - "8000"
23                - --decoder-host
24                - http://vllm-pd-with-mooncake-demo-decode-0.s-vllm-pd-with-
25                - --decoder-port
26                - "8000"
27      # 2. Prefill Worker (vLLM): Producer role
28      - name: prefill
29        replicas: 1
30        template:
31          spec:
32            volumes:
33              - name: model
34                persistentVolumeClaim:
35                  claimName: qwen2.5-7b
36              - name: dshm
```



Minor issues

submit

[Bugfix] Strengthen the check of X-data-parallel-rank in Hybrid LB mode #32314

 Merged [chaunceyjiang](#) merged 3 commits into [vllm-project:main](#) from [openanolis:dtcccc/bugfix](#)  Jan 15, 2026

report

[BugFix] Fix P/D with non-MoE DP #33037

 Merged [njhill](#) merged 3 commits into [vllm-project:main](#) from [njhill:fix-nonmoe-dp-pd](#)  on Jan 28

relate

[BugFix] Support online dense model DP without overhead #30739

 Merged [youkaichao](#) merged 7 commits into [vllm-project:main](#) from [njhill:non-moe-dp](#)  on Jan 2

[KV Connector] Add temporary, off-by-default

[VLLM_DISABLE_REQUEST_ID_RANDOMIZATION](#) workaround #34415

 Merged [vllm-bot](#) merged 5 commits into [vllm-project:main](#) from [eicherseiji:p2p-nccl-disable-request-id-randomization](#)  last month



Future plan

Core Features:

- Mooncake store integration
- Heterogeneous TP support
- PP support
- Layer-wise Transfer
- Cross-Layer KV Support
- PCP/DCP Support
- Other improvements and fixes(better error handling, physical block != config block, different hidden size per layer, HND-NHD transfer, test cases, kv events & stats, etc...)



Thanks.

